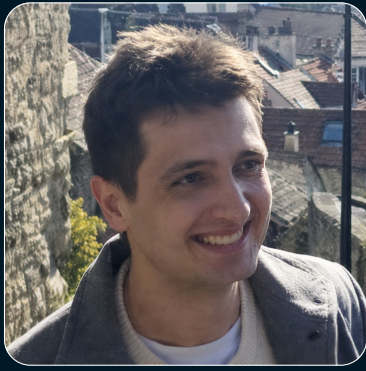




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🌐 doriangailhard.github.io

🎓 [Google Scholar](#)

🐙 [GitHub](#)

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📍 Palaiseau, France

Research Areas

Generative Modeling, Graph & Hypergraph Learning, Hierarchical Models, Flow Matching & Diffusion, Reinforcement Learning, Combinatorial Optimization

Technical Skills

ML & Scientific

PyTorch (Geometric, Distributed), JAX, NumPy

Programming

Python, C++, C

Research Infrastructure

Slurm, Hydra, Weights & Biases, Docker, Git, Linux, AWS

Academic Service

Teaching

Machine Learning with Graphs

Reviewer

NeurIPS Workshop on Graph ML, IEEE TSIPN, Learning on Graphs

Organizer

ELLIS Doctoral Symposium on AI & Sustainability

Languages

French — Native
English — C1
Spanish — B2
Italian — B2

Interests

Arts & Culture

Cinema, Art History, Humanities

Sports

Bouldering, Running, MMA

Dorian Gailhard

PhD Student in Machine Learning

I am a third-year PhD student in Artificial Intelligence at [Télécom Paris](#) (LTCI, COM-ELEC), with a defense expected in May 2027. My research focuses on scalable generative modeling and combinatorial optimization on graphs and hypergraphs, with applications to Electronic Design Automation (EDA), in collaboration with [Arteris](#).

Publications

Feature-Aware (Hyper)graph Generation via Next-Scale Prediction — ICML 2026

[Dorian Gailhard](#), Enzo Tartaglione, Lirida Naviner, Jhony H. Giraldo | [PDF](#) [Code](#)

Introduces the first hierarchical generative framework for graphs and hypergraphs that can generate both topology and features. Extends coarse-to-fine generation, previously focused on topology, to attributed structures, enabling the generation of attributed graphs and hypergraphs at scales beyond those supported by previous methods.

HYGENE: A Diffusion-Based Hypergraph Generation Method — AAAI 2025

[Dorian Gailhard](#), Enzo Tartaglione, Lirida Naviner, Jhony H. Giraldo | [PDF](#) [Code](#)

Introduces the first diffusion-based generative model for hypergraphs. Extends hierarchical graph generation to higher-order structures, providing a scalable alternative to directly modeling the exponentially large space of possible hyperedges.

Research Experience

PhD Student — May 2024 – Present

[Télécom Paris](#) — France | [Project Page](#)

Advisors: Jhony H. Giraldo, Enzo Tartaglione, Lirida Naviner

Researching scalable generative models for graphs and hypergraphs, with applications to EDA, in collaboration with Arteris. Developed hierarchical generation methods based on diffusion and flow matching, leading to first-author publications at AAAI 2025 and ICML 2026. Current work focuses on reinforcement learning for combinatorial optimization in EDA.

Research Intern — May 2023 – October 2023

[Università degli Studi di Padova](#) — Italy | [Project Page](#)

Extended continuous formulations of the Maximum Clique Problem to k -graphs. Studied Lovász extensions and continuous relaxations for discrete combinatorial optimization, proved theoretical results, and implemented the proposed methods in Python and C++.

Education

M2 MVA — Mathematics, Vision, Learning — 2022–2023

[ENS Paris-Saclay](#) — France | [Coursework & Projects](#)

Research-oriented program covering machine learning, artificial intelligence, optimization, computer vision, and applied mathematics.

Master of Science — Data Science — 2021–2023

[Eurecom](#) — France | [Coursework & Projects](#)

Engineering-oriented program covering data science, machine learning, statistical learning, cloud computing, and large-scale data processing.

Engineering Degree — 2020–2023

[Télécom Paris](#) — France | [Projects](#)

Broad engineering curriculum with a specialization in computer science, covering algorithms, software engineering, mathematics, signal processing, and telecommunications.

Preparatory Classes MPSI–MP* — 2018–2020

[Lycée Hoche](#) — France

Intensive curriculum in mathematics, physics, and computer science.